

Reachability in Two-Parametric Clock Systems

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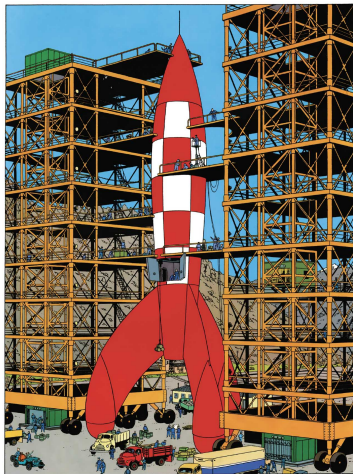
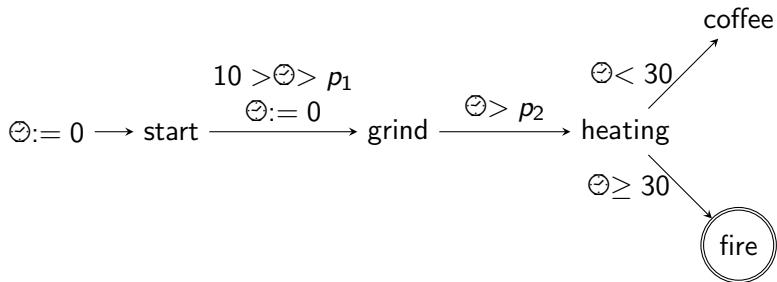
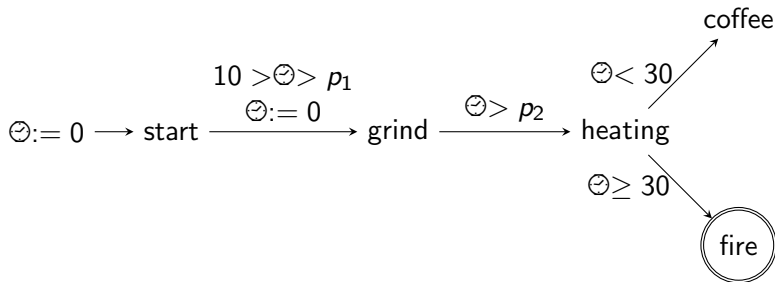


Figure: [Tintin: Destination Moon]

Parametric Timed Automata (PTA)



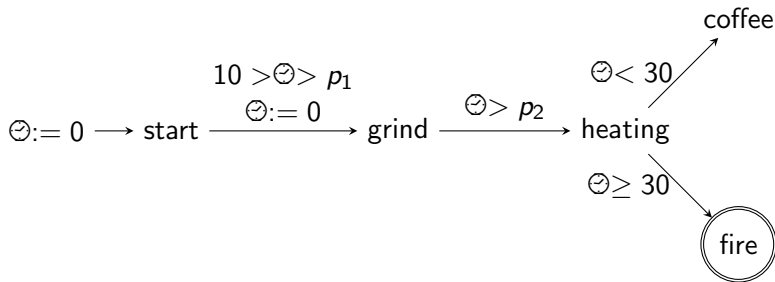
Parametric Timed Automata (PTA)



Definition

Given a (r, k) -PTA $\mathcal{A}$, does there exist a valuation of the parameters such that $\mathcal{A}$ admits an accepting run?

Parametric Timed Automata (PTA)



Definition

Given a **(2,2)-PTA** $\mathcal{A}$, does there exist a valuation of the parameters such that $\mathcal{A}$ admits an accepting run?

Params \ Clocks	1 clock	2 clocks	≥ 3 clocks
0	PSPACE-complete [AD94]		
1	PSPACE-complete [AHV93]	EXPSPACE-complete [BO17, GH20]	Undecidable [AHV93]
≥ 2		Open	

References



R. Alur and D. L. Dill.

A theory of timed automata.

Theoretical Computer Science, 126(2):183–235, 1994.



R. Alur, T. A. Henzinger, and M. Y. Vardi.

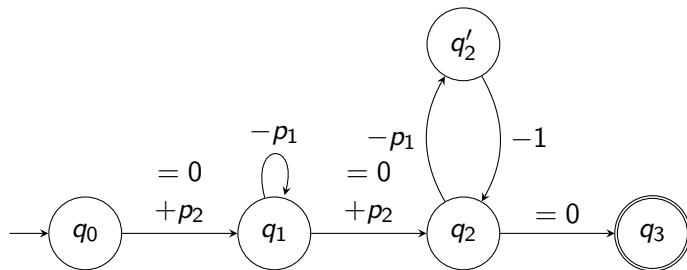
Parametric real-time reasoning.

In *Proceedings of the 25th Annual ACM Symposium on Theory of Computing (STOC)*, pages 592–601. ACM, 1993.

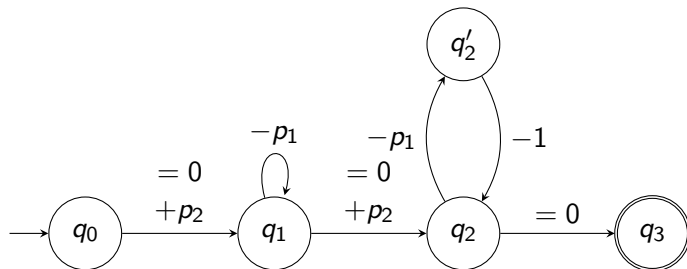


D. Bundala and J. Ouaknine.

Parametric One Counter Automata (POCA)



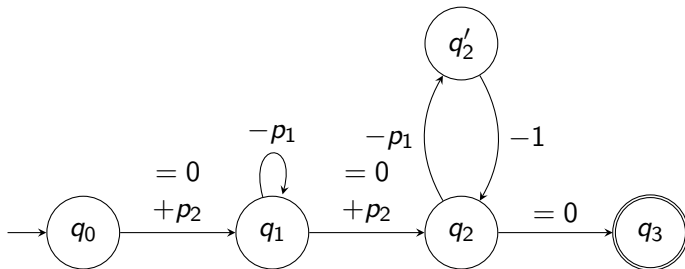
Parametric One Counter Automata (POCA)



Definition

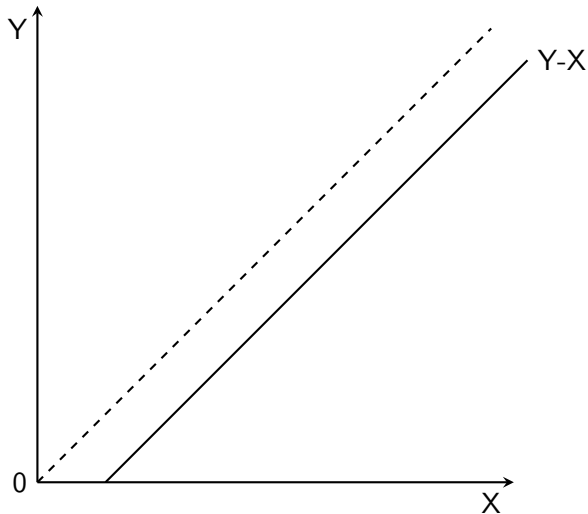
Given a k -POCA $\mathcal{C}$, does there exist a valuation of the parameters such that $\mathcal{C}$ admits an accepting run?

Parametric One Counter Automata (POCA)

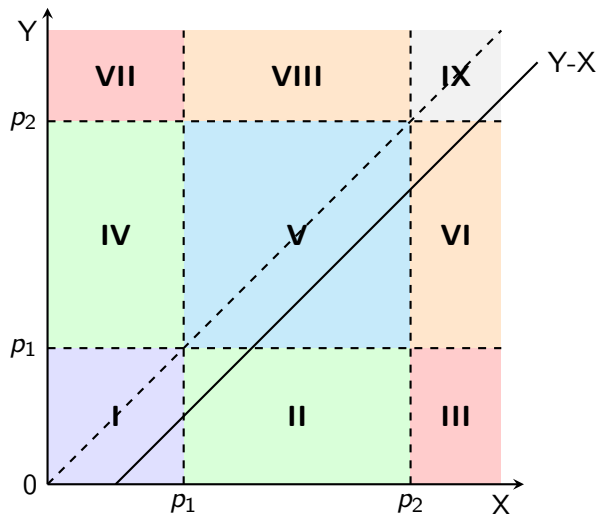


Definition

Given a **2-POCA** $\mathcal{C}$, does there exist a valuation of the parameters such that $\mathcal{C}$ admits an accepting run?



PTA to POCA



Dire que l'on va étudier seulement les POCA Est ce que l'on a besoin d'une slide pour ça

Reduction to $\exists$ PAD

$$\Phi : \exists x, y, \quad y > x \quad \vee \quad (y|x \quad \wedge \quad y + 1 \neq x)$$

Contents

- 1 Remove bad transitions
- 2 Linearly bounded by all parameters
- 3 POCA transformation
- 4 Run with $\pm p_2$ transitions
- 5 Run without $\pm p_2$ transitions

Remove " $\equiv 0 \bmod c$ " transitions

$$q \Rightarrow q, (m_1, \dots, m_r)$$

Remove " $\equiv 0 \bmod c$ " transitions

$$q \Rightarrow q, (m_1, \dots, m_r)$$

$$q \xrightarrow{+c} q' \Rightarrow q, (m_1, \dots, m_r) \xrightarrow{+c} q', (m_1 + c, \dots, m_r + c)$$

$$q \xrightarrow{+p_i} q' \Rightarrow q, (m_1, \dots, m_r) \xrightarrow{+p_i} q', (m_1 + d_{1_i}, \dots, m_r + d_{r_i})$$

$$q \xrightarrow{\bmod c_i} q' \wedge m_i = 0 \Rightarrow q, (m_1, \dots, m_r) \xrightarrow{+0} q', (m_1, \dots, m_r)$$

$$q \xrightarrow{\mathcal{G}(P, \mathbb{Z})} q' \Rightarrow q, (m_1, \dots, m_r) \xrightarrow{\mathcal{G}(P, \mathbb{Z})} q', (m_1, \dots, m_r)$$

Remove " $+ [0, p_i]$ " transitions

Lemma

We can construct an equivalent run with a finite number of $+ [0, p_i]$ between $[R, p_i - R]$.

$$\tau_1 \xrightarrow{e_1} \tau_2 \xrightarrow{e_2} \tau_3 \text{ with } e_1, e_2 \in [R, p_i - R].$$
$$\tau_1 \xrightarrow{e_1 + R} \tau_2 \xrightarrow{e_2 - R} \tau_3.$$

Linearly bounded by all parameters

$$\forall p \in P, \exists k \in \mathbb{N}, \quad counter(\pi) \leq kp$$

$$\forall p \in P, \exists k \in \mathbb{N}, \quad \text{counter}(\pi) \leq kp$$

Lemma

The emptiness problem for POCA when the counter is linearly bounded by all parameters is decidable.

$$\boxed{q, a_n p_n + \dots + a_1 p_1 + r} \Rightarrow \boxed{(q, a_n, \dots, a_1), r}$$

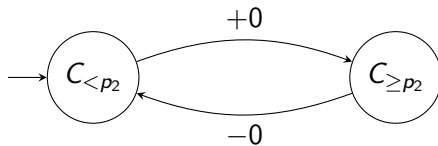
A better bound

remove $< p_2$ from C to get $C_{\geq p_2}$

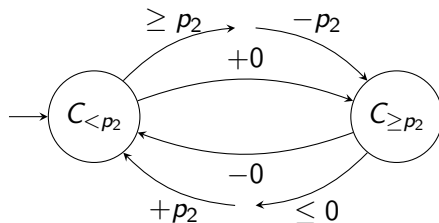
remove $\geq p_2$ from C to get $C_{< p_2}$



A better bound

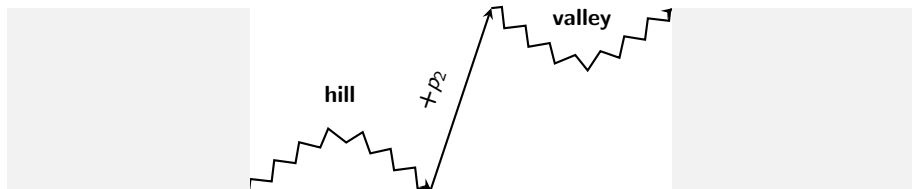
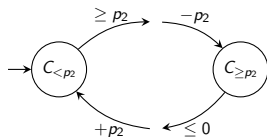


A better bound

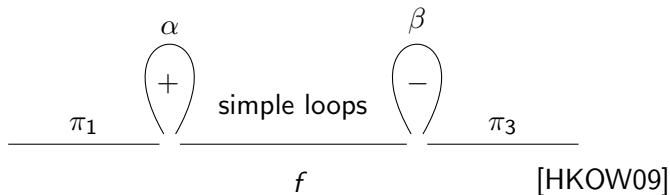
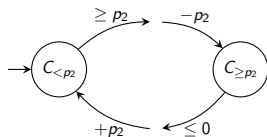


We can use $\pm p_2$ transitions in $[0, p_1]$ and $[p_2, p_2 + p_1]$.

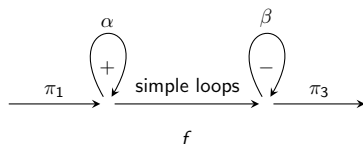
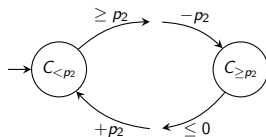
Without bridges



Without bridges



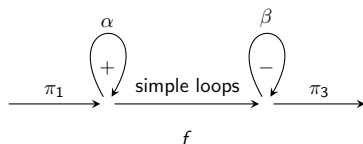
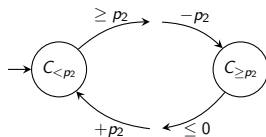
Without bridges



Lemma ([BO17] $\exists \rho \in P, \sigma \in N, \rho \times \sigma \neq 0$)

There exists an equivalent run that can be decomposed into a finite number of non-linear subruns.

Without bridges

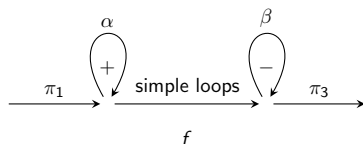
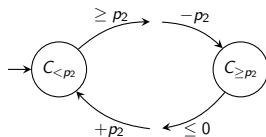


Lemma ([BO17] $\exists \rho \in P, \sigma \in N, \rho \times \sigma \neq 0$)

There exists an equivalent run that can be decomposed into a finite number of non-linear subruns.

$$\exists \vec{s}, \vec{t}, \bigwedge_{i \leq l} \Phi_{NL}(\vec{s}(i), \vec{t}(i)) \wedge \Phi(\vec{t}(i), \vec{s}(i+1))$$

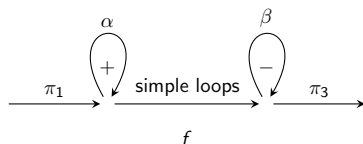
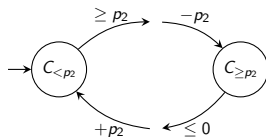
Without bridges



Lemma ([BO17] $\forall \rho \in P, \sigma \in N, \rho \times \sigma = 0$)

There exists a constant K such that there is an equivalent run in which the counter value remains bounded by $K \cdot p_1$.

Without bridges

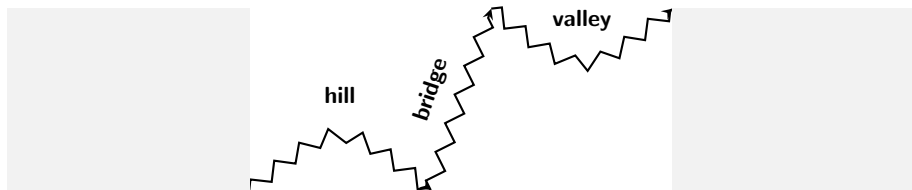
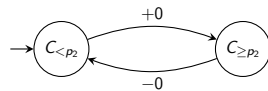


Lemma ([BO17] $\forall \rho \in P, \sigma \in N, \rho \times \sigma = 0$)

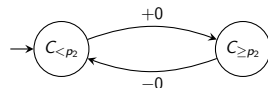
There exists a constant K such that there is an equivalent run in which the counter value remains bounded by $K \cdot p_1$.

$$\begin{array}{c}
 [p_2 - Kp_1, p_2] \\
 +2Kp_1 + 1 \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) -2Kp_1 - 1 \\
 [0, Kp_1]
 \end{array}$$

Without $\pm p_2$ transitions

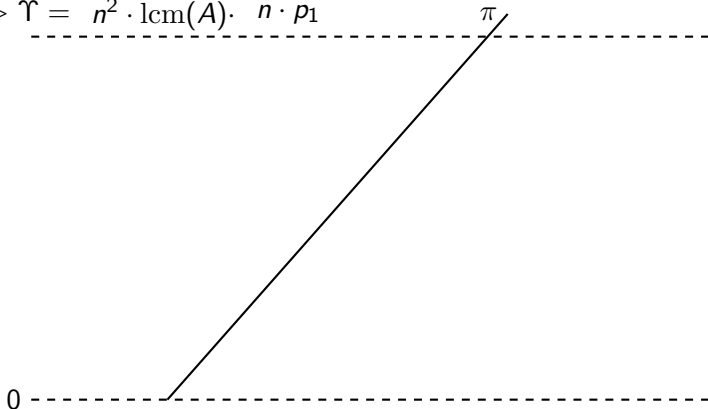


Depumping lemma

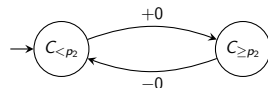


$$A = \{a \cdot p_1 + b \mid a, b \in [1, n]\}$$

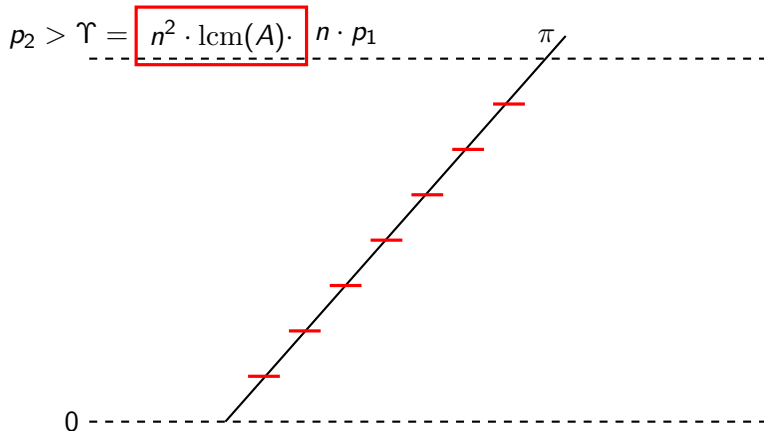
$$p_2 > \Upsilon = n^2 \cdot \text{lcm}(A) \cdot n \cdot p_1$$



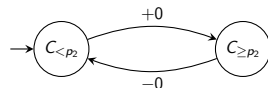
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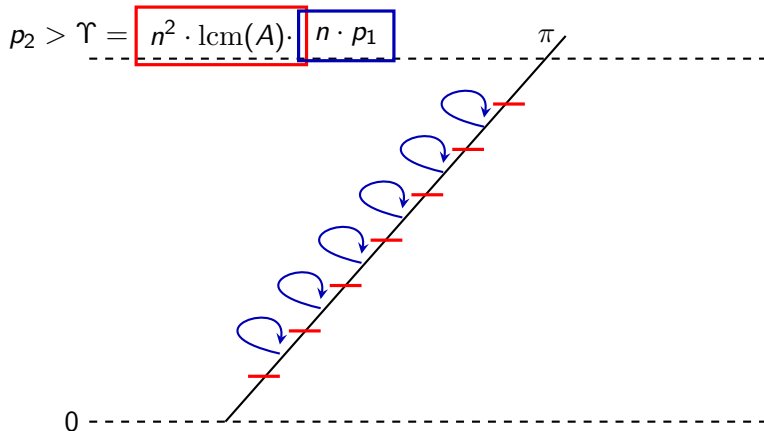
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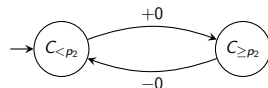
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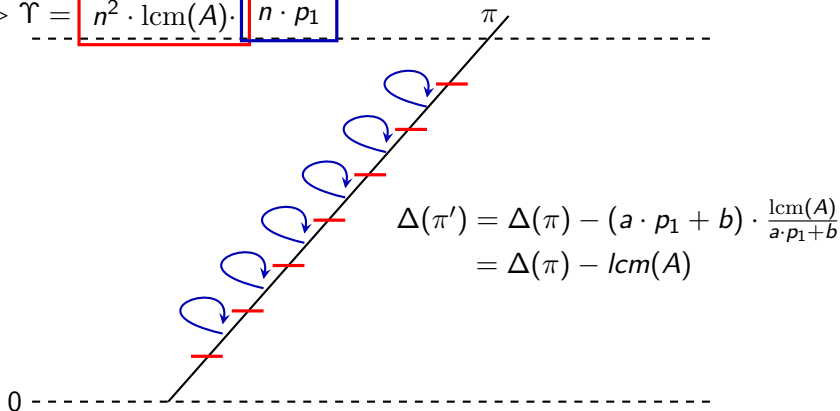


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$$p_2 > \Upsilon = \boxed{n^2 \cdot \text{lcm}(A)} \cdot \boxed{n \cdot p_1}$$



Depumping lemma

